SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned checkpoint — 19 July 2026

Authority. Corrected French lines 2888–2899; original printed p. 97; physical source-PDF p. 84; recomposed running p. 76. No `pageoriginale` token occurs inside this statement: the printed-p. 97 token occurs immediately before it, within French line 2886. Blank line 2900 is excluded. French line 2901 begins the proof and is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration. Under closed Option A, the English target's calligraphic $\mathcal{F}$ is the established English normalization from source upright F, not literal source-glyph preservation. Blackboard-bold $\mathbb{Z}$, ordinary math-italic R, and $X \setminus U$ remain explicitly disclosed target normalizations from source upright-bold $\mathbf{Z}$, upright operator R, and $X - U$.

Proposition 3.2.—Let X be a locally Noetherian prescheme, locally embeddable in a regular prescheme. Let U be an open subset of X, and let $i\colon U \to X$ be the canonical immersion. Let $n \in \mathbb{Z}$. Finally, let $\mathcal{F}$ be a coherent Cohen–Macaulay $\mathcal{O}_U$-module (on U!). The following conditions are equivalent:

(a) $R^p i_*(\mathcal{F})$ is coherent for $p < n$;

(b) for every irreducible component S' of the closure $\overline{S}$ of the support S of $\mathcal{F}$, one has

$$\operatorname{codim}(S' \cap (X \setminus U), S') > n.$$